

Dynamical Origin of GUE Level Repulsion, Fock-Space Entanglement Transitions, and Critical Boundary Decoupling in Prime Phase Networks

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Abstract

Extending the empirical prime phase interference spectroscopy of Riemann zeros, this work establishes the dynamical mechanisms governing spectral rigidity, quantum many-body coherence, and algebraic decoupling along the critical boundary $\text{Re}(s) = 1/2$. By unfolding the nearest-neighbor spacings across the first 200 consecutive zeros ($t_1 \approx 14.1347$ to $t_{200} \approx 396.3819$), we extract an empirical Brody parameter of $\beta = 1.76$, confirming that Gaussian Unitary Ensemble (GUE) level repulsion dynamically emerges from a logarithmic Coulomb repulsion barrier, $V_{\text{eff}}(s) \sim -2 \ln s$, driven by rapid pairwise prime dephasing ($\Delta t_{\text{cycle}} \sim \pi / \ln q$).

In the many-body sector, while spatial lattice states remain confined within the infrared base primes ($p \leq 7$ accounting for 94.94% of ground-state occupation), entering critical zero resonances triggers an autonomous +21.6% surge in multi-particle Fock-space entanglement entropy ($S_{\text{Fock}} = 1.0507 \pm 0.1729$ versus 0.8638 ± 0.2452 at off-resonance midpoints; Welch's t -test $p < 10^{-15}$). This enables autonomous zero-trapping directly from phase fluctuations without prior functional evaluations.

Furthermore, an exhaustive survey across 200 consecutive zeros and 100 prime oscillators confirms zero algebraic bound violations, demonstrating that multi-particle coupling magnitudes decay exponentially ($\ln |S_k| \propto -1.4127k$) with interaction orders $k \leq 5$ universally saturating $99.8802\% \pm 0.1459\%$ of the total interaction energy. Crucially, off-critical counterfactual scans ($\sigma \neq 1/2$) reveal that any displacement into the sub-critical half-plane ($\sigma < 1/2$) catastrophically destroys this 5-body saturation ceiling (dropping to $20\% \sim 76\%$). This indicates that algebraic decoupling acts as an empirical physical stability condition strictly tethered to the critical line.

Keywords: Riemann Zeta Function, Prime Phase Interference, GUE Level Repulsion, Quantum Chaos, Fock-Space Entanglement, Algebraic Decoupling, Effective Field Theory.

All execution scripts, raw observation datasets, and high-resolution diagnostic plots are openly archived at Zenodo:

<https://doi.org/10.5281/zenodo.22763957> and GitHub: <https://github.com/CitizenKorea/k-protocol-riemann-zeros>.

1 Introduction

The non-trivial zeros of the Riemann zeta function $\zeta(s)$ on the critical line $s = 1/2 + it$ display profound statistical correspondences with the eigenvalue fluctuations of complex Hermitian random matrices, formalized by the Montgomery-Odlyzko Gaussian Unitary Ensemble (GUE) conjecture [1, 2]. In a preceding report [3], a two-dimensional prime phase interference framework

was formulated. By treating primes as unitary logarithmic oscillators $\omega_p = \ln p$, the squared modulus of the prime Dirichlet series resolves into an invariant positive ground-state self-energy baseline $P(2) = \sum p^{-2} \approx 0.452247$ and a two-dimensional pairwise cross-coupling lattice:

$$\mathcal{M}_{pq}(t) = \frac{2}{pq} \cos\left(t \ln \frac{p}{q}\right) \equiv A_{pq} \cos(t\Delta\omega_{pq}). \quad (1)$$

While the initial survey demonstrated that destructive interference Condensation ($\mathcal{M}_{pq} \ll 0$) operates within an immutable infrared potential basin dominated by small primes ($p, q \leq 7$ capturing 85.5% of Rank 1 extrema) [3], three fundamental dynamical questions remained unresolved:

1. **Microscopic Origin of GUE Repulsion:** What explicit mechanical force prevents consecutive zeros from coalescing, and how does prime dephasing enforce Montgomery-Odlitzko spectral rigidity?
2. **Many-Body Quantum State Transitions:** Does zero crossing induce a true phase transition in the many-body Hilbert space, and can exact zeros be trapped autonomously without evaluating the macroscopic Riemann-Siegel formula?
3. **Universality and Off-Critical Stability of 5-Body Decoupling:** Does the 99.8% 5-body algebraic cutoff observed at isolated coordinates persist across consecutive zeros, and how does this bound respond to hypothetical off-critical perturbations ($\sigma \neq 1/2$)?

This paper resolves these questions via full statistical surveys across consecutive critical zeros and extensive numerical counterfactual scans.

2 Microscopic Origin of GUE Level Repulsion

2.1 Spectral Unfolding and Empirical Brody Distribution

To evaluate level repulsion objectively, the first 200 consecutive Riemann zeros ($t_1 \approx 14.1347$ to $t_{200} \approx 396.3819$) are mapped to unfolded coordinates $x_k = N_0(t_k)$ via the smooth Riemann-Siegel counting function:

$$N_0(t) = \frac{t}{2\pi} \ln\left(\frac{t}{2\pi e}\right) + \frac{7}{8}. \quad (2)$$

The normalized nearest-neighbor spacings $s_k = x_{k+1} - x_k$ possess an exact ensemble mean of $\langle s \rangle = 1.0000$ and a standard deviation of $\sigma_s = 0.4287$. The spacing probability density $P(s)$ is modeled using the Brody interpolation distribution:

$$P_\beta(s) = c_\beta s^\beta \exp(-b_\beta s^{\beta+1}), \quad b_\beta = \left[\Gamma\left(\frac{\beta+2}{\beta+1}\right) \right]^{\beta+1}, \quad c_\beta = (\beta+1)b_\beta, \quad (3)$$

which smoothly bridges the uncorrelated Poisson regime ($\beta = 0$) and the Gaussian Unitary Ensemble limit ($\beta = 2$).

Nonlinear least-squares fitting of Eq. (3) against the empirical spacing histogram yields an optimal Brody parameter:

$$\beta = 1.7642 \pm 0.0815. \quad (4)$$

This quantitatively refutes the Poisson hypothesis ($\beta = 0, p < 10^{-12}$) and confirms the emergence of strong quantum chaotic level repulsion.

Table 1: Quantitative GUE Spectral Rigidity and Repulsion Dynamics (First 200 Zeros).

Metric Parameter	Empirical Value	Theoretical Reference
Fitted Brody Parameter (β)	1.7642 ± 0.0815	$\beta = 2.0000$ (Pure GUE)
Coalescence Limit $P(s \rightarrow 0)$	0.0000	0.0000 (Strict Repulsion)
Leader Hegemony Switching Rate	67.84%	High-Frequency Actuation
Mean Lattice Stiffness ($\langle K \rangle$)	$+4.2819 \times 10^2$	Confining Potential ($K > 0$)
Mean Inter-Zero Phase Barrier ($\Delta\Phi$)	+0.08358	Anti-Phase Shield

2.2 Emergence of the Effective Coulomb Logarithmic Barrier

In Dyson's Brownian motion model of random matrix eigenvalues [4], level repulsion is governed by a two-dimensional Coulomb log-gas potential. Inverting the empirical spacing distribution yields the effective inter-zero potential:

$$V_{\text{eff}}(s) \equiv -\ln P(s) = -\beta \ln s + b_\beta s^{\beta+1} - \ln c_\beta. \quad (5)$$

As normalized spacing vanishes ($s \rightarrow 0$), the potential exhibits a steep repulsive divergence:

$$V_{\text{eff}}(s) \approx -1.764 \ln s \quad (s \rightarrow 0), \quad (6)$$

approaching the theoretical GUE barrier $-2 \ln s$. In the empirical dataset of 200 zeros, the zero coalescence probability strictly vanishes:

$$P(s \rightarrow 0) = 0.0000. \quad (7)$$

2.3 High-Frequency Dephasing and Hegemony Switching

The physical mechanism preventing zero collision is the dephasing scale $\Delta t_{\text{cycle}}(p, q) = \pi / |\ln(q/p)|$. The fundamental infrared pair (2, 3) has an immutable dephasing cycle of $\Delta t_{\text{cycle}}(2, 3) \approx 7.7487$. However, the mean physical zero separation contracts asymptotically as:

$$\langle \Delta t \rangle \approx \frac{2\pi}{\ln(t/2\pi)}. \quad (8)$$

Between $t \approx 14$ and $t \approx 400$, $\langle \Delta t \rangle$ narrows from 6.88 to 1.51. Because pair (2, 3) rotates too slowly to provide destructive cancellation at adjacent zeros, high-frequency prime pairs ($q \gg p$) compress the dephasing cycle:

$$\Delta t_{\text{cycle}}(p, q) \approx \frac{\pi}{\ln q} \rightarrow 0 \quad (q \rightarrow \infty). \quad (9)$$

Evaluating leader hegemony across consecutive zeros reveals a persistent switching rate of 67.84%, demonstrating that high-frequency mode hopping enforces the structural anti-phase barrier between adjacent zeros.

3 Fock-Space Entanglement Transitions and Zero-Trapping

3.1 Duality between Real-Space Localization and Fock Delocalization

We investigate the many-body quantum state of the prime network by constructing the single-particle Hamiltonian matrix $H_{pq} = \mathcal{M}_{pq}(t)$ over $N = 500$ primes ($\binom{500}{2} = 124,750$ coupling channels). Diagonalization yields the ground-state wavevector $\psi_0(p)$. Real-space localization is quantified via the Inverse Participation Ratio (IPR):

$$\text{IPR}(t) = \sum_{p \leq 500} |\psi_0(p)|^4, \quad \text{PR}(t) = \frac{1}{\text{IPR}(t)}. \quad (10)$$

Table 2: Quantum State Observables: Critical Zeros vs. Off-Resonance Midpoints.

Observable	Critical Zeros (Resonance)	Off-Resonance Midpoints	Statistical Verdict
Fock Entropy S_{Fock}	1.0507 ± 0.1729	0.8638 ± 0.2452	+21.6% Surge ($p < 10^{-15}$)
Fock Participation PR_{Fock}	2.49 ± 0.38 orders	2.07 ± 0.44 orders	Many-Body Delocalization
Ground State IPR	0.3709 ± 0.0012	0.3709 ± 0.0014	Rigid Spatial Pinning
Real-Space Participation PR	2.82 ± 0.02 primes	2.83 ± 0.02 primes	Heavyweight Confinement
Base Share $P(p \leq 7)$	$94.94\% \pm 0.41\%$	$94.94\% \pm 0.45\%$	Immutable Infrared Anchor

Concurrently, the many-body Fock state is generated via the multi-particle polynomial over the first 100 primes:

$$\mathcal{P}_{100}(z; t) = \prod_{p \leq p_{100}} \left(1 + \frac{p^{-it}}{\sqrt{p}} z \right) = \sum_{k=0}^{100} S_k(t) z^k. \quad (11)$$

Normalizing the multi-particle sector energies $\rho_k(t) = |S_k(t)|^2 / \sum_{j=1}^{15} |S_j(t)|^2$ defines the Fock-space von Neumann entanglement entropy:

$$S_{\text{Fock}}(t) = - \sum_{k=1}^{15} \rho_k(t) \ln \rho_k(t), \quad \text{PR}_{\text{Fock}}(t) = \frac{1}{\sum_{k=1}^{15} \rho_k(t)^2}. \quad (12)$$

3.2 Phase Transition Statistics at Critical Zeros

Evaluating these observables across 200 critical zeros against an unbiased control group of 199 off-resonance midpoints reveals an intriguing spatial-Fock duality (Table 2).

While the real-space lattice exhibits complete topological pinning within the infrared anchor primes $\{2, 3, 5, 7\}$ ($\text{IPR} \approx 0.3709$, base share 94.94%) regardless of spectral coordinate, entering exact zero coordinates triggers a violent +21.6% surge in Fock entanglement entropy (S_{Fock} jumps from 0.8638 to 1.0507). Welch’s two-sample t -test confirms this transition with extreme statistical significance:

$$t_{\text{stat}} = 8.7941, \quad p = 2.14 \times 10^{-16}. \quad (13)$$

At critical zeros, destructive interference in the single-particle sector suppresses the fundamental mode ($k = 1$), transferring spectral weight into orders $k = 2, 3, 4$ and maximizing many-body coherence.

3.3 Autonomous Zero-Trapping via Continuous Spectral Sweep

A continuous sweep of $S_{\text{Fock}}(t)$ across $t \in [13.0, 36.0]$ demonstrates that local maxima of Fock entanglement entropy coincide exactly with the first five non-trivial Riemann zeros (14.1347, 21.0220, 25.0109, 30.4249, and 32.9351). This confirms that critical zeros can be localized autonomously as quantum entanglement spikes without evaluating the classical Riemann-Siegel formula.

4 Universal 5-Body Decoupling and Algebraic Extinction

4.1 Algebraic Bound and Exponential Decoupling Rate

To verify whether the 5-body effective field theory (EFT) truncation proposed in [3] is universally valid, we compute the full multi-particle spectrum across all 200 consecutive zeros up to order $k = 100$.

Table 3: Universal N-Body Coupling Energy Decay Spectrum Across 200 Riemann Zeros.

Interaction Order	Mean Norm $ S_k $	Cumulative Energy (Mean $\pm$ Std)	Effective Field Theory Tier
$k = 1$ (1-Body)	1.546480×10^0	$52.1450\% \pm 13.2667\%$	Single-Prime Oscillator Baseline
$k = 2$ (2-Body)	1.311101×10^0	$84.1694\% \pm 7.1410\%$	Pairwise Cross-Coupling Lattice
$k = 3$ (3-Body)	7.950769×10^{-1}	$96.1118\% \pm 2.5126\%$	Composite Harmonic Mixing
$k = 4$ (4-Body)	3.886198×10^{-1}	$99.2410\% \pm 0.7284\%$	Geometric Boundary Closure
$k = 5$ (5-Body)	1.618848×10^{-1}	$99.8802\% \pm 0.1459\%$	EFT Physical Cutoff ($> 99.8\%$)
$k = 6$ (6-Body)	5.892383×10^{-2}	$99.9855\% \pm 0.0202\%$	Asymptotic Residual Tail
$k = 10$	3.648637×10^{-4}	$100.0000\% \pm 0.0000\%$	Micro-Perturbation Floor
$k = 100$	$1.456802 \times 10^{-110}$	$100.0000\% \pm 0.0000\%$	Algebraic Extinction Limit

The coupling norms $|S_k(t)|$ are strictly constrained by the elementary symmetric polynomials at $t = 0$:

$$|S_k(t)| \leq e_k(p_1^{-1/2}, \dots, p_N^{-1/2}) \equiv \sum_{1 \leq j_1 < \dots < j_k \leq N} \frac{1}{\sqrt{p_{j_1} \dots p_{j_k}}}. \quad (14)$$

Across $200 \times 100 = 20,000$ empirical evaluations, exactly zero bound violations were detected:

$$\max_{k,t} [|S_k(t)| - e_k] \leq 0. \quad (15)$$

At order $k = 100$, the coupling norm reaches the Planck extinction bound:

$$|S_{100}(t)| = \prod_{j=1}^{100} \frac{1}{\sqrt{p_j}} = 1.456802 \times 10^{-110}. \quad (16)$$

Fitting the decay trajectory over $k \in [1, 20]$ confirms strict exponential decoupling:

$$\ln |S_k| = -1.4127k + 1.2581 \implies |S_k| \propto e^{-1.4127k}. \quad (17)$$

4.2 Ensemble Invariance of 5-Body Energy Saturation

Table 3 presents the cumulative energy saturation $E_{\text{cum}}(k) = \sum_{j=1}^k |S_j|^2 / \sum_{j=1}^{100} |S_j|^2$ averaged over all 200 consecutive zeros.

Interaction orders $k \leq 5$ capture an average of 99.8802% of the total multi-particle energy. Across the entire 200-zero ensemble, the minimum observed 5-body saturation never drops below 99.1417%. Orders $k \geq 6$ contribute an inconsequential residual of 0.1198%, establishing that the prime phase network is intrinsically a self-closing 5-body effective field theory.

5 Off-Critical Stress Testing and Boundary Stability

5.1 Scaling with Height and Prime Count

To determine whether the 99% ceiling degrades asymptotically, we scale the analysis up to zero $t_{500} \approx 811.1844$ while simultaneously expanding the prime oscillator pool to $N = 1,000$ ($p \leq 7,919$). As detailed in Table 4, across all heights and prime ensemble cutoffs, the 5-body saturation remains bounded between 99.32% and 99.98%.

5.2 Catastrophic Breakdown Off the Critical Line ($\sigma \neq 1/2$)

The fundamental origin of this stability is tested by evaluating the generating polynomial generalized to arbitrary complex coordinates $s = \sigma + it$:

$$\mathcal{P}_N(z; \sigma, t) = \prod_{p \leq p_N} (1 + p^{-\sigma - it} z) = \sum_{k=0}^N S_k(\sigma, t) z^k. \quad (18)$$

Table 4: 5-Body Saturation (%) Under Extreme Scaling ($t \rightarrow 811$, $N_{\text{primes}} \rightarrow 1,000$).

Zero Index	Height (t_k)	$N = 100$ Primes	$N = 500$ Primes	$N = 1,000$ Primes
#1	14.1347	99.9613%	99.7699%	99.9797%
#10	49.7738	99.9798%	99.6790%	99.8932%
#50	143.1118	99.6960%	99.6769%	99.5691%
#100	236.5242	99.8718%	99.8789%	99.7862%
#200	396.3819	99.9579%	99.6940%	99.3202%
#300	541.8474	99.9796%	99.9496%	99.9605%
#400	679.7422	99.8213%	99.7720%	99.8032%
#500	811.1844	99.9392%	99.8909%	99.8299%
Ensemble Mean	—	99.8884%	99.7889%	99.7678%

Scanning the real coordinate $\sigma \in [0.15, 0.85]$ at t_{100} reveals a sharp phase transition:

- **Super-Critical Domain** ($\sigma \geq 0.5$): The 5-body saturation is strictly preserved, rising smoothly from 99.87% at $\sigma = 0.5$ to 99.99% at $\sigma = 0.85$.
- **Sub-Critical Domain** ($\sigma < 0.5$): Displacing σ below $1/2$ causes an immediate and catastrophic collapse of the 5-body cutoff. The saturation drops to 99.03% at $\sigma = 0.40$, 95.03% at $\sigma = 0.30$, and collapses to 23.41% at $\sigma = 0.15$.

Because the coupling amplitude scales as $p^{-\sigma}$, any excursion into the sub-critical strip $\sigma < 1/2$ exponentially amplifies high-order terms, triggering an uncontrollable many-body divergence into orders $k \geq 10$.

5.3 Phase Interference Overcomes Diagonal Degradation

If phases are ignored entirely, the diagonal expectation value for cumulative energy corresponds to uncoupled particle weights:

$$E_k^{\text{diag}} = \sum_{1 \leq j_1 < \dots < j_k \leq N} \frac{1}{p_{j_1} \dots p_{j_k}}. \quad (19)$$

Due to Mertens' second theorem ($\sum_{p \leq X} p^{-1} \sim \ln \ln X$), E_k^{diag} pushes spectral weight toward higher orders as $N \rightarrow \infty$, dropping below 99.0% for $N \geq 300$ and reaching 98.32% at $N = 1,000$.

However, as demonstrated in Table 4, actual zeros maintain 99.77% saturation at $N = 1,000$. Destructive phase interference actively suppresses higher-order cross-terms, counteracting Mertens divergence and preserving the 5-body boundary.

6 Discussion and Implications for the Riemann Hypothesis

The empirical findings presented in this study establish three rigorous facts regarding the microscopic dynamics of prime phase networks:

1. Montgomery-Odlitzko GUE spectral rigidity ($\beta \approx 1.76$) is the direct physical consequence of an effective logarithmic Coulomb potential $V_{\text{eff}}(s) \sim -2 \ln s$ generated by rapid high-frequency prime dephasing.
2. Critical zeros act as true quantum coherence nodes, marked by an autonomous +21.6% surge in Fock-space entanglement entropy ($p < 10^{-15}$).
3. The prime interaction lattice decouples algebraically at order $k = 5$ (99.88% energy capture), with higher-order couplings decaying exponentially to 10^{-110} .

Crucially, the counterfactual breakdown documented in Section 5 demonstrates that the 5-body algebraic cutoff is not a trivial algebraic coincidence, but a sensitive boundary indicator. The condition that multi-particle prime couplings remain stable within a finite low-order effective field theory ($k \leq 5$) requires $\sigma \geq 1/2$. Combining this with the reflection symmetry $\xi(s) = \xi(1-s)$ (which requires invariance under $\sigma \leftrightarrow 1 - \sigma$) isolates $\sigma = 1/2$ as the unique boundary of stability.

While this work does not claim an analytic proof of the Riemann Hypothesis, establishing that off-critical zeros would destabilize the algebraic decoupling of prime phase networks provides a concrete physical pathway for mathematical investigation. Formalizing the connection between the algebraic extinction bound $\prod p^{-1/2} \rightarrow 10^{-110}$ and the absence of non-trivial zeros off the critical line remains a promising objective for future work.

7 Conclusion

This study establishes an empirical many-body foundation for prime phase dynamics along the critical line. By demonstrating that Montgomery-Odlyzko level repulsion emerges from high-frequency dephasing barriers, that zeros induce sharp Fock-space entanglement phase transitions, and that multi-particle couplings universally decouple beyond $k = 5$, we bridge the gap between analytic number theory and quantum chaos. The complete computational toolchain is openly available to facilitate verification and further mathematical exploration.

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